

S. Shayan Daneshvar

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Education

Jan 2023– March 2025 MSc, Computer Science, University of Manitoba, GPA: **4.5/4.5**, **Supervisor:** Dr. Shaowei Wang
Thesis Title: Exploring Representation-level Augmentation and RAG-based Vulnerability Augmentation with LLMs for Vulnerability Detection
Courses: Advanced Data Mining, Deep Learning with CNNs, Data-driven Software Engineering Research Methodologies, Image-based Generative Methods in Machine Learning (all **4.5/4.5 - A+**).

Sep 2018– June 2022 BSc, Computer Engineering, K.N. Toosi University of Technology,
GPA: **19.21/20** (US CGPA: **4/4, A+**), Ranked **1st** in class of **76**
Thesis Title: Reflection Removal of In-vehicle Images (19.5/20) **Supervisor:** Dr. Nasihatkon
Selected Courses: Computer Vision, Natural Language and Speech Processing, Database Design, Advanced Programming, Algorithm Design, Differential Equations, Compiler Design, Microprocessors & AVR Assembly, Discrete Mathematics, Hardware & Software Co-design, Numerical Methods, Software Engineering II, Internet Engineering. (all grades **20/20**)
Software Engineering I (19.9/20), Linear Algebra (19.9/20), Operating Systems (19.75/20), Theory of Languages & Automata (19.75/20), Engineering Mathematics (19.5/20).

Fields of Interests

- Generative AI, Large Language Models, Agentic AI, Deep Learning, Computer Vision & Image Processing
- Data Generation and Synthesis, Automated Software Engineering, AI4SE, SE4AI, MLOps

Honors and Awards

Feb 2025 **Faculty of Graduate Studies Research Completion Award**, C\$5000, University of Manitoba.

Jan 2023– Jan 2025 **University of Manitoba Graduate Fellowship (UMGF)**, C\$28000, University of Manitoba.

Jan 2023– Jan 2024 **International Graduate Student Entrance Scholarship (IGSES)**, C\$6500, University of Manitoba.

Aug 2022 Straight MSc. in Artificial Intelligence **Admission Offer**, Sharif University of Technology.

Sep 2019– July 2020 **Dean's List Inclusion**, Computer Engineering Faculty's Dean, K. N. Toosi University of Technology.

Aug 2018 Ranked within the **Top 1%** Nationwide University Entrance Exam.

Work Experience

Nov 2025– present **LLM Research Engineer**, Huawei Canada – *Agentic Software Engineering*

April 2025– September 2025 **Machine Learning Engineer**, MacDon Industries Ltd. – *Unsupervised Domain Adaptation (Contract via Zoomi Technologies)*

Nov 2024– March 2025 **Machine Learning Engineer Intern**, MacDon Industries Ltd.
Generative AI for Digital Agriculture (Mitacs Accelerate)

May 2022– Dec 2022 **Senior Software Engineer and Developer**, Mahsan Co. – *Distributed Data-Analytics Platform*

June 2022– Nov 2022 **Instructor**, Mahsan Co. – *Spring Cloud, Spring Framework, and Microservices Architecture*

Nov 2021– May 2022 **Backend Software Engineer and Consultant**, Tosan Soha – *E-Wallet & Trading Platforms*

May 2021– Nov 2021 **Java Backend Developer**, Pinket – *Chat & Shopping Services*

Jan 2020– May 2021 **Java Instructor and Mentor (Part-time)**, Mapsa HR and Training

June 2019– Sep 2019 **Intern**, Mapsa HR and Training – *Backend Web Development*

Academic Experience

Jan 2023– March 2025	Graduate Research Assistant, ML4SE , University of Manitoba, Supervisor: Dr. Shaowei Wang
Sep 2023– Apr 2025	Grader/TA, Software Engineering II , University of Manitoba, Instructor: Dr. Shaowei Wang
Sep 2024– Dec 2024	TA, Applied Computational Intelligence , University of Manitoba, Supervisor: Dr. Ahmed Ashraf
Jan 2024– Apr 2024	Grader, Machine Learning , University of Manitoba, Instructor: Timothy Zapp
May 2023– Apr 2024	Grader, Software Engineering I , University of Manitoba, Instructor: Robert Guderian
Sep 2023– Dec 2023	Grader, Project Management , University of Manitoba, Instructor: Dr. Shaowei Wang
July 2021– July 2022	Undergraduate Research Assistant, Computer Vision , KNTU, Supervisor: Dr. Behrooz Nasihatkon
Sep 2021– Feb 2022	Teaching Assistant, Database Design , KNTU, Instructor: Dr. Saeed Farzi
Feb 2021– July 2021	Head Teaching Assistant, Operating Systems , KNTU, Instructor: Dr. Hamed Khanmirza
Feb 2021– July 2021	Head Teaching Assistant, Algorithm Design , KNTU, Instructor: Dr. Ali Ahmadi
Sep 2019– Aug 2020	Head Teaching Assistant, Advanced Programming , KNTU, Instructor: Mehdi Zamanian

Technical Skills

Languages	Java, Python, C#, Kotlin, C/C++, Javascript & Typescript, Scala.
Concepts	OOP, GoF Design Patterns, Architectural Styles and Patterns, AOP, Prompt Engineering, RAG, SQL/NewSQL & NoSQL Databases, Concurrency and Parallelism, Event-Driven Programming, TDD/BDD, Web Development, Microservices Architecture, Android Development, Agentic AI Deep Learning, Generative Models, LLMs, Transformers, CNNs, RNN and LSTM, Computer Vision and Image Processing, AI for Software Engineering, Generative AI, Agile Software Development & Project Management (Scrum & Kanban)
ML Frameworks	PyTorch, Tensorflow, Keras, Megatron-LM, HuggingFace, OpenCV, TensorBoard, vLLM.
Web Frameworks	Spring Framework(MVC, Webflux, Cloud, Security, Data, Boot), Angular 2+.
Tools	LaTeX, Git, Gitlab, Docker, Nginx, Kafka, Hibernate, Flyway, Kubernetes.
IDEs	IntelliJ IDEA, Pycharm, Visual Studio, VS Code, Claude Code, Codex.
Databases	MySQL, Postgres, YugabyteDB, MS Server, MongoDB, Cassandra, SQLite.
Other Libraries	Hibernate, JUnit, Mockito, , JavaFX & WPF, Maven.
Operating Systems	Windows, Ubuntu Linux.

Publications

Citations: 70+ - h-index: 3

SS. Daneshvar, Y. Nong, X. Yang, S. Wang, and H. Cai, "VulScribeR: Exploring RAG-based Vulnerability Augmentation with LLMs," TOSEM journal, 2026. [DOI](#). [Q1, IF: 7.0].

SS. Daneshvar, D. Tan, S. Wang, and C. Leung, "A Study on Mixup-Inspired Augmentation Methods for Software Vulnerability Detection," EASE Conference, 2025. [DOI](#). [CORE A].

N. Rezaei, **SS. Daneshvar**, B. Nasihatkon, S. Seidi, M. Rezazadeh, "The application of barcode readable assay and linear regression RGB analysis using a customized smartphone app in on-chip electromembrane extraction for simultaneous determination of heavy metal ions," Microchemical Journal, Volume 197, 2024. [DOI](#). Related to the **Visual Concentration Estimation** project. [Q1, IF: 5.4]

SS. Daneshvar and S. Wang, "GUI Element Detection Using SOTA YOLO Deep Learning Models," [preprint](#), 2024.

Language

Farsi (Native), English (Advanced Proficiency - C1), Arabic (A2), French (A1)

Mar 2025– Mar 2027 **CELP-IP-General (English)**, Total: **11** | Reading: 11 | Listening: 11 | Speaking: 11 | Writing: 11

Sep 2021– Sep 2023 **TOEFL iBT (English)**, Total: **100** | Reading: 22 | Listening: 30 | Speaking: 22 | Writing: 26

Selected Projects

Important Coursework Projects

Feb - Apr 2024	Single Image Reflection Removal with MAMBA Role: Researcher Image-based Generative Methods in ML Professor: Dr. Chris Henry Replicated the best SOTA Single Image Reflection Removal model (DSRNet), created an image-based version of a SOTA state-space model (Mamba/S6), and replaced attention-based modules of DSRNet with Mamba modules to improve the performance of the model. I also investigated the effect of the Cosine Annealing learning rate schedule and AdamW's weight decay.
Oct - Dec 2023	Exploring Vulnerability Augmentation with Mixup-based Representation-level Techniques Role: Developer & Researcher Advanced Data Mining Professor: Dr. Carson K. Leung Used various SenMixUp-inspired methods to augment vulnerable code samples, after transforming them into a vector format and before training. I also developed a vulnerability-preserving heuristic to maintain the vulnerable state of code samples.
Mar - Apr 2023	Brain Tumor Segmentation and Survival Rate Prediction with 3DUNet variants Role: Developer & Researcher Deep Learning with CNNs Professor: Dr. Ahmed Ashraf 3D Segmentation of Brain Tumors on BraTS2020 dataset. I created, trained, and tested 3 variants of 3D UNet, namely Vanilla 3D UNet, Residual 3D UNet, and 3D UNet with a custom Attention mechanism. I also used the segmentation results to train an FCN to predict survival rates.
Jan - Mar 2023	GUI Element Detection using SOTA YOLO models Role: Researcher Data-driven Software Engineering Professor: Dr. Shaowei Wang An empirical study on GUI Element Detection with the latest small-sized YOLO models (5, 6, 7, and 8). Specifically, I compared SOTA YOLO models for detecting Android GUI elements and showed that Text Views are often misclassified while Drawers and Switches are easier to detect.
June - July 2021	BSc Thesis, Reflection Removal of In-vehicle Images (with UNets) Role: Developer & Researcher BSc Thesis Professor: Dr. Behrooz Nasihatkon I built and trained two variations of UNet to remove or decrease the effect of the reflections of in-vehicle images such that the resulting picture would look natural. For the dataset, I used a dataset of real-world streets and road images (CamVid) to synthesize a new dataset with reflections of various objects.
June - July 2021	Soccer Player Detection & Classification Role: Developer Computer Vision Professor: Dr. Behrooz Nasihatkon Detection & Classification of players & referees from a recorded soccer match & visualization of their location on the soccer field. I used the KNN background subtraction algorithm with basic morphology techniques for the detection task. I labeled over 3K pictures for the classification task and trained a CNN. I also used perspective transform to merge 3 cameras' videos and map coordinates to a flat birds-eye view field.
Oct 2020–Feb 2021	Dapixi.ir Role: Product Owner & Developer Software Engineering Professor: Dr. Mehdi Esnaashari Dapixi is a free photo sharing website similar to Pinterest. Responsibilities: back-end web development in Java, minor bug fixes in the Angular front-end, and development of the content-based & collaborative filtering photo recommender systems in Python.
Apr – July 2020	Database & Web Application Design of a Website Role: Developer Database Design Professor: Dr. Saeed Farzi . Database Design & Fullstack web development of a website similar to divar.ir.
July 2021 – July 2022	Other Major Projects Visual Concentration Estimation Role: Android & Python Developer, Researcher Supervisor: Dr. Behrooz Nasihatkon Detection of linear and QR barcodes including chemicals with CNNs, concentration estimation using linear and log-linear regressors, and outlier detection with PCA.
June – Sep 2021	Chibaladi.com Role: Developer, Project Manager Backend design & development of the Chibaladi website. Chibaladi was a website for assessing developers' knowledge and recommending videos for closing their knowledge gaps.

Academic Service

2026–present Reviewer, **ACM Transactions on Software Engineering and Methodology (TOSEM)**
 2026 Participant, **ICSE Shadow Program Committee (Shadow PC)**

Certificates

August 2025 ACM Professional Member | ACM.
 April 2025 Mitacs Accelerate | Mitacs.
 September 2021 From Shallow to Deep Learning | ISMVIP.
 August 2021 Fundamentals of Reinforcement Learning | IEEE KNTU Student Branch.
 October 2019 Java Web Developer Certificate | Javacup Association.
 Mar 2019 – present Over 25 other certificates in various fields, mostly in Software Engineering | LinkedIn Learning.

Hobbies

Playing Electric Guitar, Weightlifting, Running, Hiking, Playing Tennis, and Reading

References

Dr. Shaowei Wang Associate Professor, Department of Computer Science, University of Manitoba.
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Dr. Behrooz Nasihatkon Assistant Professor, Department of Computer Engineering, K. N. Toosi University of Technology.
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Dr. Christopher Henry Associate Professor, Department of Computer Science, University of Manitoba.
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Dr. Haipeng Cai Associate Professor, Department of Computer Science and Engineering, University at Buffalo.
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Dr. Carson Leung Full Professor, Department of Computer Science, University of Manitoba.
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Dr. Shayan A. Tabrizi Tech Lead, Software Architect and Manager, Mahsan Co., Tehran.
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Dr. Ahmed Ashraf Associate Professor, Department of Electrical and Computer Engineering, University of Manitoba.
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Dr. Babak Nasersharif Associate Professor, Department of Computer Engineering, K. N. Toosi University of Technology.
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